

Towards an Edge Cloud Based Coordination Platform for Multi-User AR Applications Built on Open-Source SLAMs

Balázs Sonkoly^{1,2,3*}, Bálint György Nagy^{1,2}, János Dóka^{1,2}, Zsófia Kecskés-Solymosi^{1,2}, János Czentye^{1,2,3},
Bence Formanek⁴, Dávid Jocha⁴, Balázs Péter Gerő⁴

¹HSN Lab, Department of Telecommunications and Media Informatics, Faculty of Electrical Engineering and Informatics,
Budapest University of Technology and Economics, ²MTA-BME Network Softwarization Research Group,

³ELKH-BME Cloud Applications Research Group, ⁴Ericsson Research

ABSTRACT

Multi-user and collaborative AR applications pose several challenges. The expected user experience requires accurate pose information for each device and precise synchronization of the respective coordinate systems in real-time. Unlike mobile phones or AR glasses running on battery with constrained resource capacity, cloud and edge platforms can provide the computing power for the core functions under the hood. In this paper, we propose a novel edge cloud based platform for multi-user AR applications realizing an essential coordination service among the users. The latency critical, computation intensive Simultaneous Localization And Mapping (SLAM) function is offloaded from the device to the edge cloud infrastructure.

Index Terms: Human-centered computing—Visualization—Visualization design and evaluation methods

1 INTRODUCTION

Extended Reality (XR) applications, including Augmented Reality (AR), Mixed Reality (MR) and Virtual Reality (VR), were launched from science fiction and have already landed in “reality”. These applications can reshape our society enabling novel services, new ways of interactions and immersive experiences in many fields. Online gaming, Industry 4.0, healthcare, architecture are just highlighted examples with special importance and business potential.

One of the key components mainly affecting the user experience is the AR or VR device. The other essential constituents are the special purpose, compute-intensive functions, running in the background, which enable the real-time operation. Several providers operate cloud based remote rendering platforms (e.g., NVIDIA GeForce Now, PlayStation Now) and the customers can rent the resources on demand instead of buying expensive hardware devices. Furthermore, we can make more steps towards more utilization of cloud platforms. Other functions can also be offloaded to the cloud if the end-to-end latency is controlled carefully and kept within a predefined bound. These networking requirements are (or will be) fulfilled by 5G and 6G systems or a new generation of WiFi networks.

In this study, we propose a novel edge cloud based AR platform for running distributed, multi-user AR applications. We investigate the benefits and the limits of the approach and evaluate the performance of a proof-of-concept prototype which offloads the Simultaneous Localization And Mapping (SLAM), which is responsible for the real-time localization of the AR device. We address open-source SLAM libraries making use of different algorithms. Multiple implementations and devices are evaluated in order to answer the question: can we provide the required level of user experience for AR users (even for AR gamers) making use of edge cloud based SLAMs?

*sonkoly.balazs@vik.bme.hu

2 PROPOSED ARCHITECTURE

We propose an edge cloud based solution, where the camera images and sensor data are streamed towards a dedicated coordination service from AR devices. The SLAM function is offloaded from the device and arbitrary algorithms can be invoked in the remote environment. In order to enable the coordination among the users, the per-device coordinate systems have to be synchronized which requires to build and use a joint global map. This is a core task of the coordination service and the details are hidden from the users. Making use of the global map, novel AR applications can be realized that are not possible with the current technologies. The architecture of the proposed system is depicted in Fig. 1. On the devices, camera images and IMU data are collected and sent to the coordination service. Since the user experience is greatly affected by the response time of the SLAM function, it is crucial to have a proper network connection between the devices and the edge cloud infrastructure.

At the cloud side, each device has its own service instance including one or multiple SLAM modules and a selector component. We run several SLAM algorithms side by side, so that we can choose the best regarding mapping and tracking results. This is done by the SLAM selector module, which decides which algorithm to use based on the camera image data and the availability of pose estimations.

3 IMPLEMENTATION

We have implemented a proof-of-concept prototype [1]. As it is shown in Fig. 1, the client application is built on the Unity 3D engine. We have implemented a native library in C++ to stream the camera images and IMU data, and addressed two different mobile platforms, namely, Android and iOS platforms, with slightly different implementation details. Our coordination service has been implemented on top of the Robot Operating System (ROS) [4] and the components are packaged into distinct software containers. The global map is currently built on ROVIOLI and maplab [5], and the map is built based on explicitly merged dedicated missions. Later, additional algorithms and map handlers can be added.

In the current configuration, we have incorporated two different open-source SLAM libraries: ROVIOLI [5] and LSD-SLAM [2]. Additional implementations can easily be added later. The output of the SLAMs are monitored by the module is called SLAM Image Analyzer (SIA) and the “better” (or faster) pose response is selected for transmission towards the device. For the SIA module, different types of neural networks were used.

4 MEASUREMENT AND EVALUATION

We use a robotic arm (Universal Robot 5E) to move the AR device on preliminary defined 3D trajectories. The real position is captured from the robot software and it can be used as a ground truth value.

As it is shown in Fig. 3, the average runtime of ROVIOLI, including the queueing delays, is above 80 ms with large variance, while LSD-SLAM exhibits much better performance calculating the pose within 7.5 ms in all scenarios. In case of Android, the encoding time of a frame takes around 72 ms due to the library limitation,

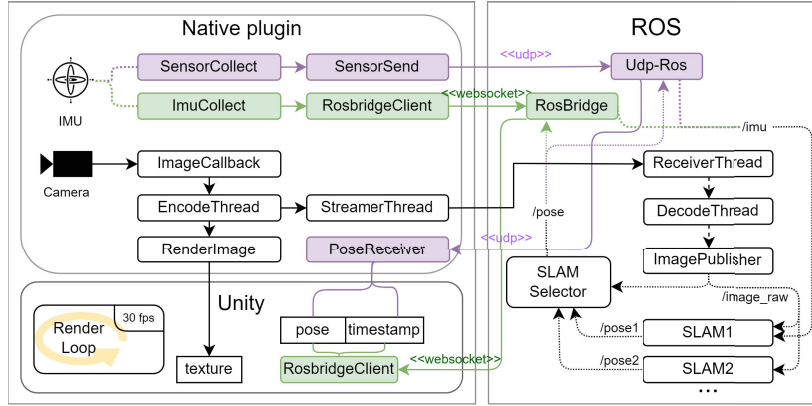


Figure 1: Proof-of-concept prototype. Clients implemented on Android and iOS (left); Coordination platform (right).

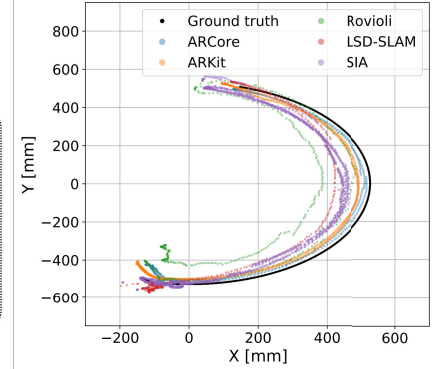


Figure 2: The performance of different SLAMs using a simple trajectory (top view only).

Table 1: ATE and RPE for different SLAMs [m]

	ARCore	ARKit	Rovioli	LSD	SIA
ATE	0.104	0.104	0.198	0.220	0.198
RPE	0.008	0.007	0.013	0.014	0.012

therefore the E2E delay in all scenarios is above 100 ms, which unfortunately has an impact on the experience. The results with iOS are promising, especially together with LSD-SLAM. In the edge setup, the average E2E latency is kept around 50 ms which means that the pose information can be updated within 2-frame delay for most of the frames (assuming 30 FPS frame rate).

Besides the strict latency characteristics which is essential for the real-time operation, accuracy is the other dimension which directly impacts the user experience. The results of the robotic accuracy measurements are shown in Fig. 2. The black curve indicates the ground truth, and as baselines, the results of ARCore (Android) and ARKit (iOS) are also depicted. Besides single SLAMs, we show the results of our ensemble module (SIA) combining ROVIOLI and LSD-SLAM. ARCore and ARKit outperform the open-source methods, however, the errors are not extreme and in certain applications, this performance can be accepted (position coordinates are in millimeters). We also calculated quantitative metrics, shown in Tab. 1. The Relative Pose Error (RPE) shows the drift of the trajectory over a fixed time interval and the Absolute Trajectory Error (ATE) measures the global difference compared to the baseline trajectory [3]. LSD-SLAM shows better performance than ROVIOLI, while the ensemble solution is able to outperform both SLAMs in terms of ATE. However, RPE indicates performance issues for SIA which can be the result of sudden changes in pose values at switching events.

To sum up, the revealed and experienced accuracy is acceptable for simple games, however it is currently not enough for critical applications. If we target high-quality AR experiences, the coordination service has to be combined with local enhancements (e.g., sophisticated predictions or filtering mechanisms). But we think that the platform can be extended towards this direction.

5 CONCLUSION

In this paper, we have proposed a novel edge cloud based coordination platform for multi-user AR applications. We have focused on the latency sensitive and computation intensive SLAM function which was offloaded from the device to the edge cloud infrastructure. We have defined a dedicated measurement methodology providing the necessary ground truth information. Following the methodology, we analyzed the latency characteristics and the accuracy of the platform

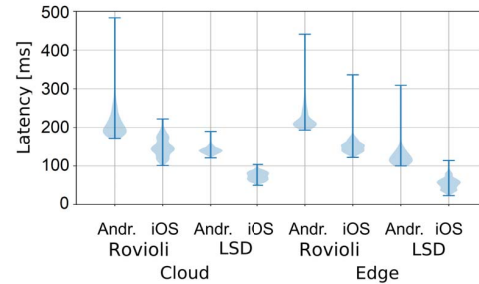


Figure 3: End-to-end latency statistics

via real experiments. The current version of the proof-of-concept prototype provides sufficient services for simple AR applications, however further improvements and additional components are required to enable high-quality AR experiences.

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